

Centrifugal Pump Impeller

Structural FEA and Modal Analysis

ANSYS Mechanical Static Structural and Modal Simulation

Materials: Structural Steel, Aluminum Alloy, Titanium Alloy

Operating Speeds: 3000, 6000, 9000 RPM

Jorge Lopez | University of Houston | May 2026

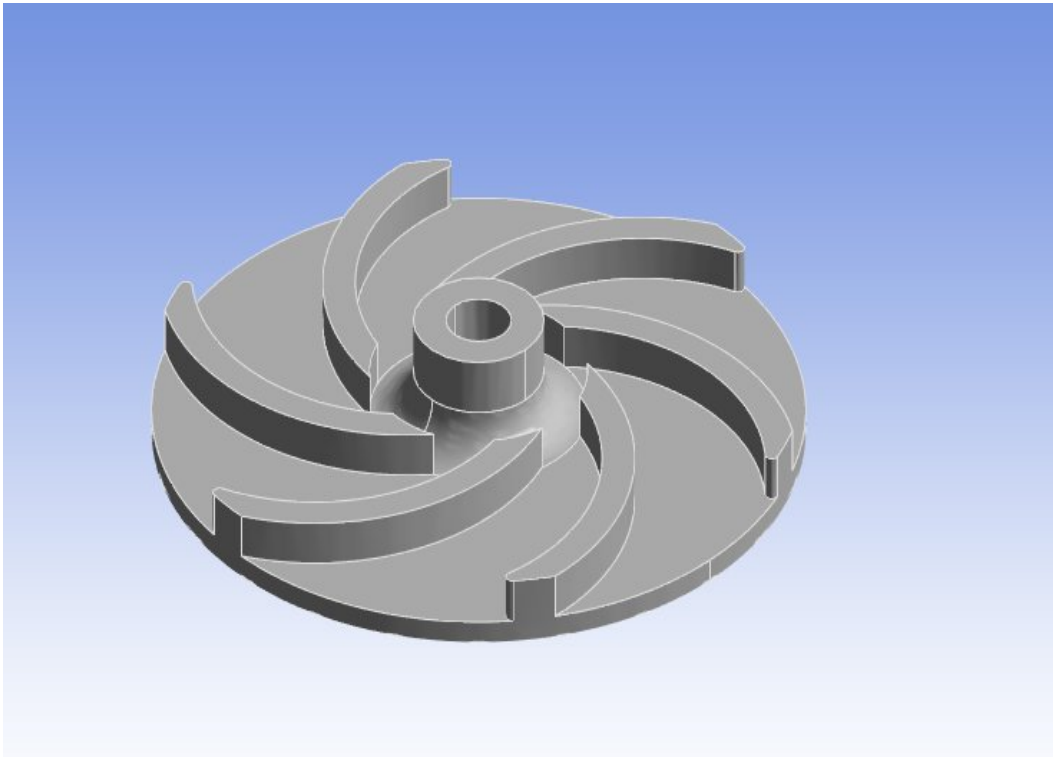


Figure 1. Centrifugal pump impeller geometry modeled in Autodesk Inventor and imported into ANSYS Mechanical.

1. Introduction

Centrifugal pump impellers are among the most mechanically demanding components in oil and gas operations. Electric submersible pumps used by companies like Baker Hughes operate continuously downhole at elevated speeds, and impeller failure can cost hundreds of thousands of dollars in lost production and retrieval operations. Understanding how material selection and rotational speed interact to determine structural integrity is therefore a core design concern for rotating equipment engineers.

This project uses ANSYS Mechanical to perform static structural analysis and modal analysis on a centrifugal pump impeller geometry across three engineering materials and multiple operating speeds. The goal is to identify the failure threshold for each material, compare fatigue life and deformation behavior, and determine whether any natural frequencies fall within the operating range where resonance could be a concern.

2. Geometry

The impeller was designed from scratch in Autodesk Inventor using a sketch-based approach. The geometry consists of twelve backward-swept blades mounted on a circular base plate with a central hub and shaft bore. Key dimensions are as follows: impeller outer radius 150 mm, hub outer radius 30 mm, shaft bore radius 12 mm, blade height 20 mm, and base plate thickness 10 mm. The backward-swept blade profile is consistent with standard centrifugal pump design practice for minimizing flow separation at design flow rate.

The completed geometry was exported as a STEP file and imported into ANSYS Mechanical for meshing and analysis.

3. Materials

Three materials were evaluated to understand the trade-offs between strength, weight, and fatigue performance at elevated rotational speeds.

Property	Structural Steel	Aluminum Alloy	Titanium Alloy
Density (kg/m3)	7850	2770	4620
Young's Modulus (GPa)	200	71	96
Yield Strength (MPa)	250	280	930
Poisson's Ratio	0.30	0.33	0.36

4. Boundary Conditions

Two boundary conditions were applied for the static structural analysis. A fixed support was applied to the inner cylindrical surface of the shaft bore, representing the impeller mounted on a rigid shaft. Rotational velocity was then applied about the Y axis at three speeds: 3000, 6000, and 9000 RPM, corresponding to 314.16, 628.32, and 942.48 rad/s respectively. These speeds bracket the typical operating range of electric submersible pumps used in oil production, which commonly run between 3000 and 4500 RPM.

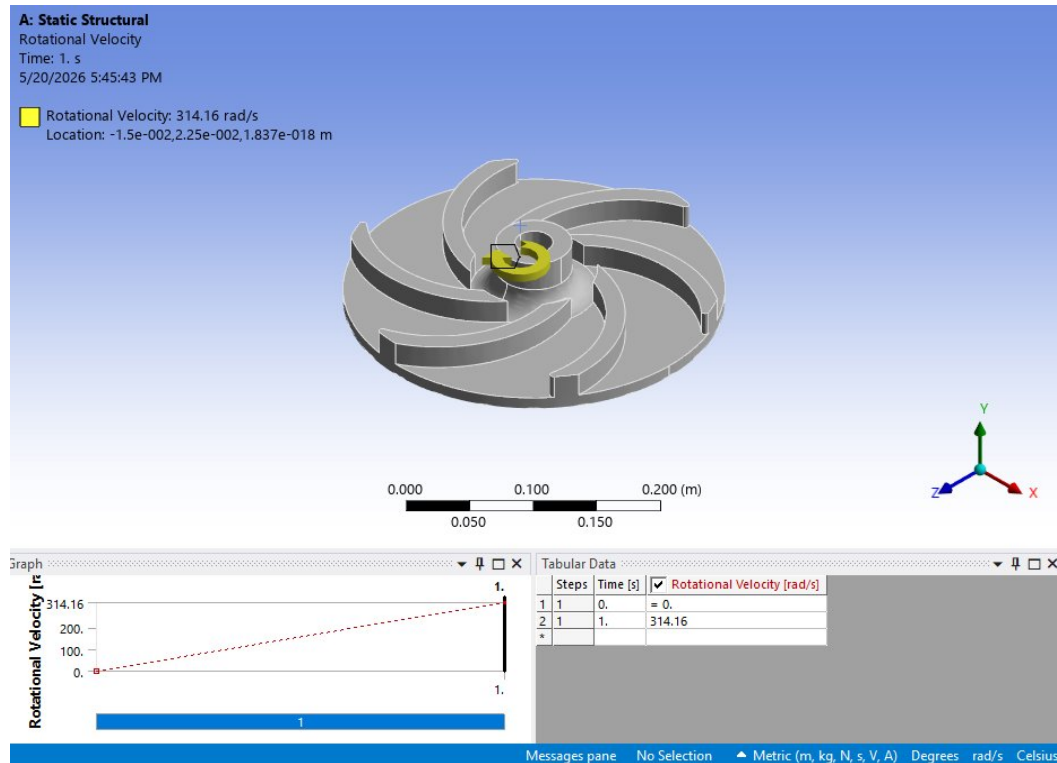


Figure 2. Rotational velocity boundary condition applied at 3000 RPM (314.16 rad/s) about the Y axis.

5. Mesh

An unstructured tetrahedral mesh was generated using ANSYS Mechanical's automatic meshing with adaptive sizing. The mesh was refined near blade surfaces and the hub fillet regions where stress gradients are steepest. The final mesh provided adequate resolution to capture the centrifugal stress distribution across the blade span while keeping solve times reasonable on a desktop workstation.

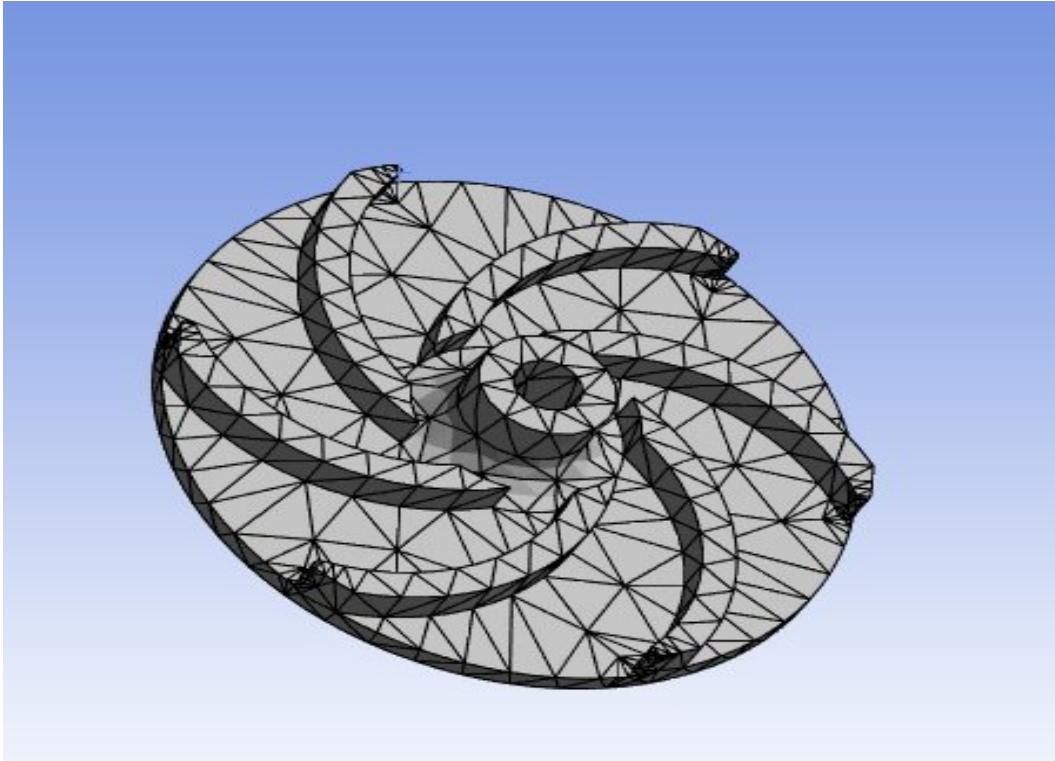


Figure 3. Finite element mesh showing tetrahedral elements with refinement near blade roots and hub.

6. Static Structural Results

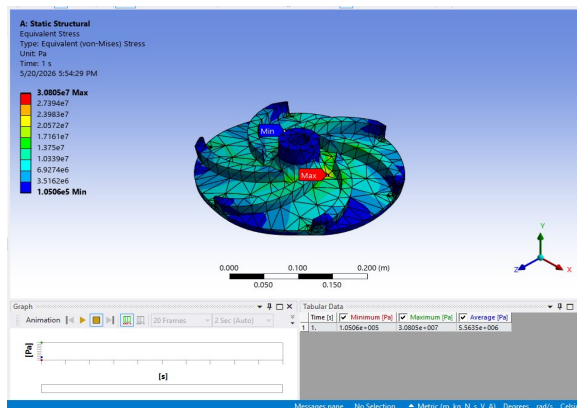
Static structural analysis was performed for Steel at all three speeds and for Aluminum and Titanium at 9000 RPM. The following results were extracted for each case: equivalent von Mises stress, total deformation, minimum safety factor, fatigue damage, and fatigue life in cycles.

Material	RPM	Max Stress (MPa)	Max Deform. (mm)	Min Safety Factor	Min Life (cycles)	Status
Steel	3000	30.8	0.034	2.80	1,000,000+	Safe
Steel	6000	109.6	0.135	0.70	150,000	Fail
Steel	9000	277.3	0.303	0.31	8,396	Fail
Aluminum	9000	97.7	0.367	0.85	22,000,000+	Marginal
Titanium	9000	162.6	0.367	N/A	N/A	See note

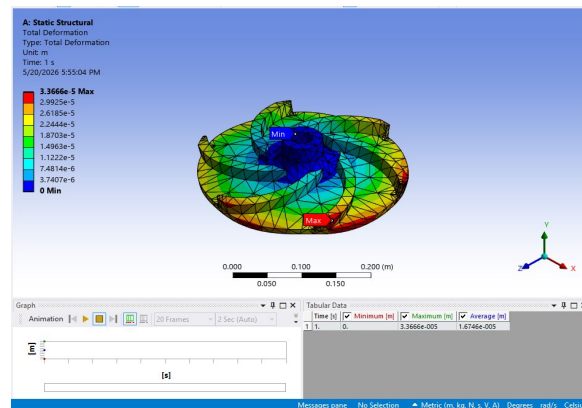
Note: Titanium safety factor and fatigue life were not extracted in this study. Stress and deformation results are available for comparison purposes.

6.1 Steel Results

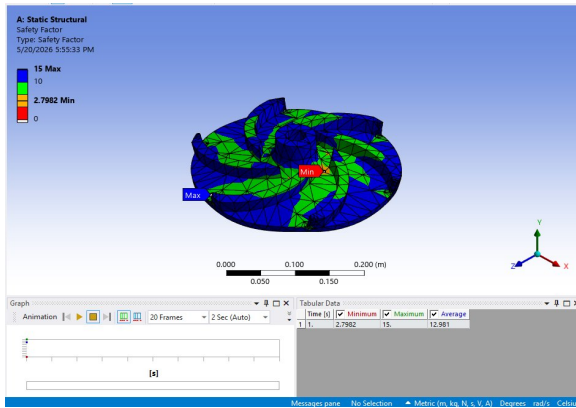
3000 RPM



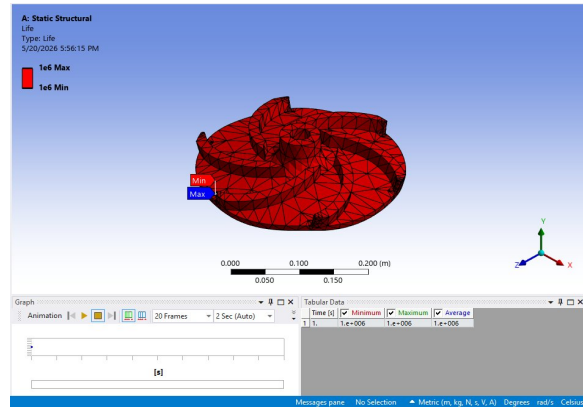
Von Mises Stress (max 30.8 MPa)



Total Deformation (max 0.034 mm)



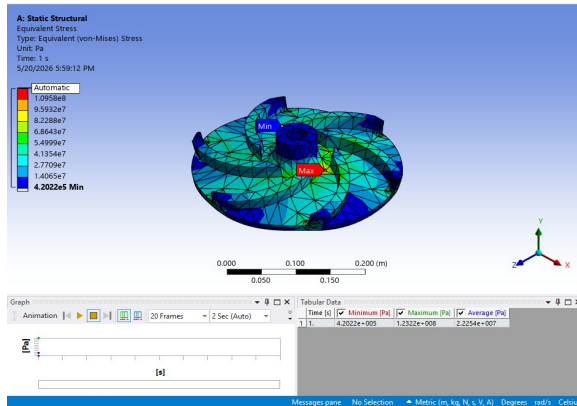
Safety Factor (min 2.80)



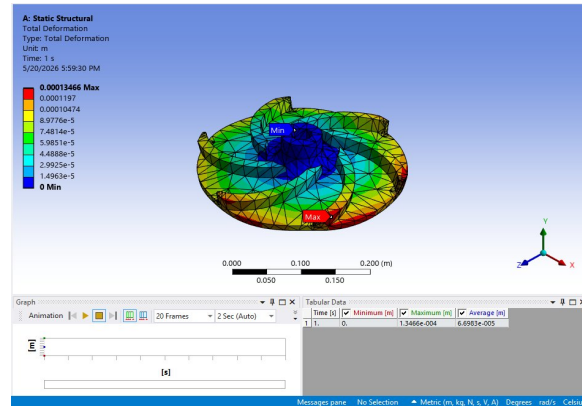
Fatigue Life (min 1e6 cycles)

Figure 4. Steel impeller at 3000 RPM. Maximum stress is well below yield strength with safety factor of 2.80.

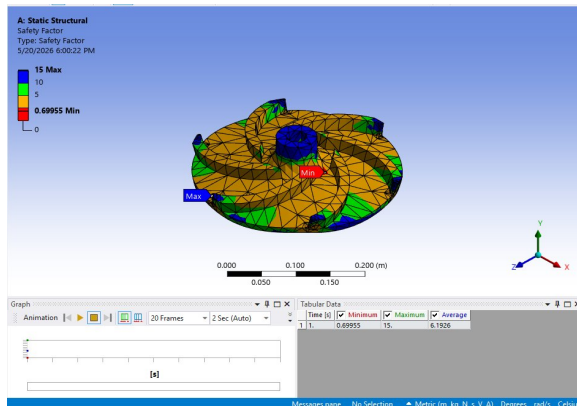
6000 RPM



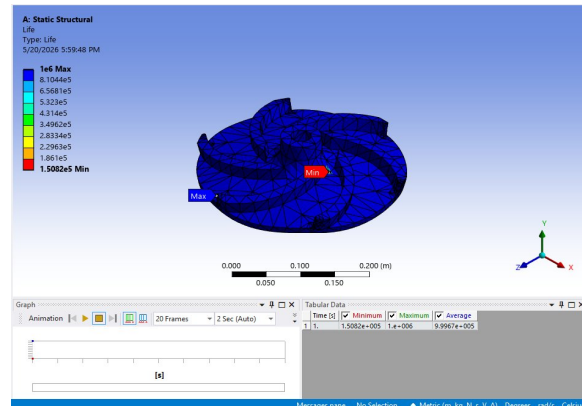
Von Mises Stress (max 109.6 MPa)



Total Deformation (max 0.135 mm)



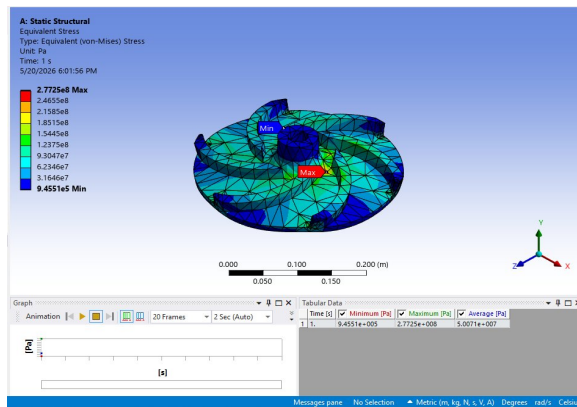
Safety Factor (min 0.70)



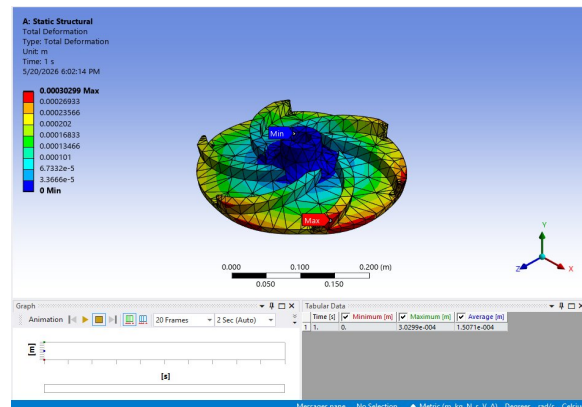
Fatigue Life (min 1.5e5 cycles)

Figure 5. Steel impeller at 6000 RPM. Safety factor drops below 1.0, indicating onset of fatigue failure.

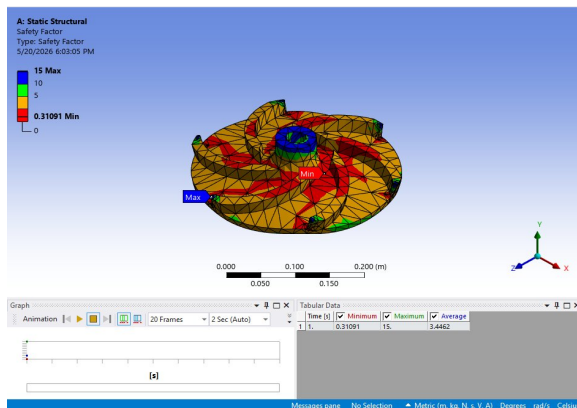
9000 RPM



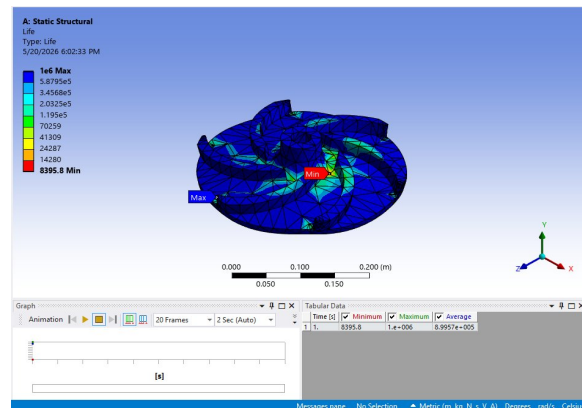
Von Mises Stress (max 277.3 MPa)



Total Deformation (max 0.303 mm)



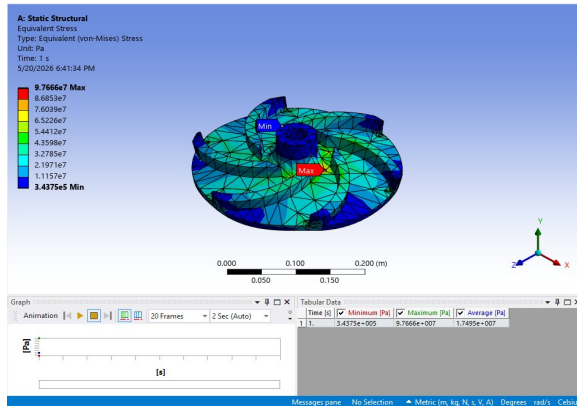
Safety Factor (min 0.31)



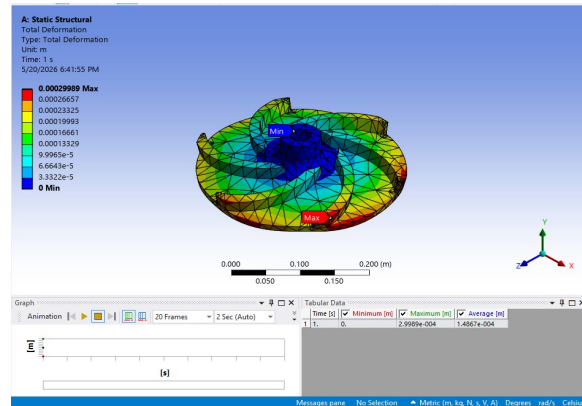
Fatigue Life (min 8,396 cycles)

Figure 6. Steel impeller at 9000 RPM. Stress exceeds yield strength in blade root regions. Critical failure expected.

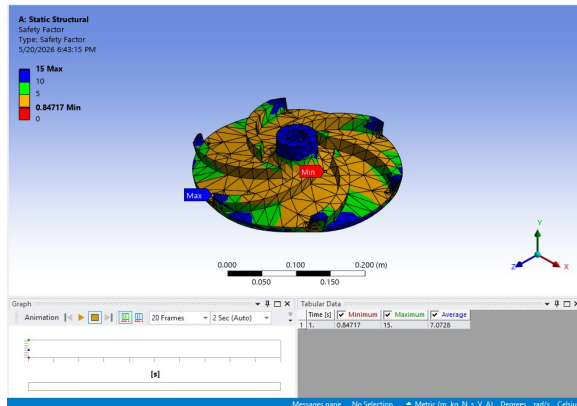
6.2 Aluminum Alloy at 9000 RPM



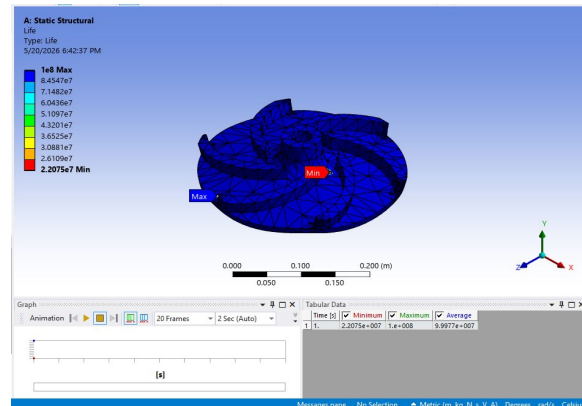
Von Mises Stress (max 97.7 MPa)



Total Deformation (max 0.367 mm)



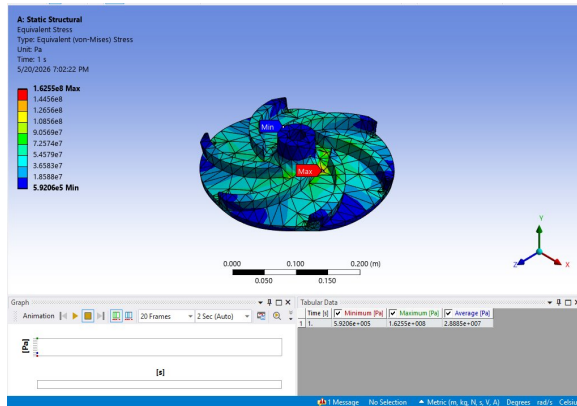
Safety Factor (min 0.85)



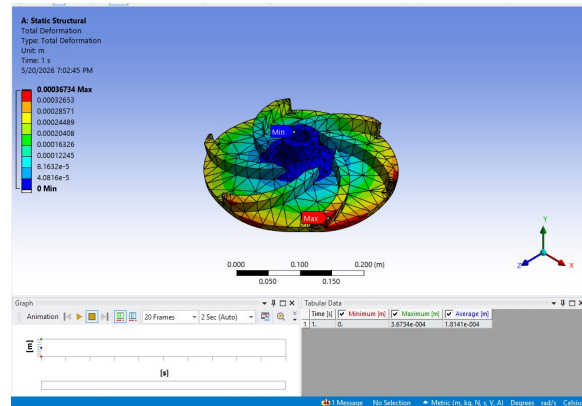
Fatigue Life (min 2.2e7 cycles)

Figure 7. Aluminum impeller at 9000 RPM. Lower stress than steel due to lower density, but safety factor still below 1.0.

6.3 Titanium Alloy at 9000 RPM



Von Mises Stress (max 162.6 MPa)



Total Deformation (max 0.367 mm)

Figure 8. Titanium impeller at 9000 RPM. Stress is intermediate between steel and aluminum. Higher yield strength of 930 MPa means this material has the most margin at elevated speeds.

7. Modal Analysis

Modal analysis was performed on all three materials to extract the first six natural frequencies. This is important because if a natural frequency falls near the blade passing frequency or shaft excitation frequency at operating speed, resonance can amplify vibration and accelerate fatigue failure dramatically. For a 12-blade impeller running at 3000 RPM, the blade passing frequency is 600 Hz. Any natural frequency near this value would be a design concern.

Mode	Aluminum (Hz)	Steel (Hz)	Titanium (Hz)
1	643.67	699.60	707.67
2	652.14	709.88	717.54
3	713.64	774.07	783.84
4	821.65	910.82	912.56
5	825.02	915.77	916.94
6	1073.10	1211.90	1203.30

All three materials show first natural frequencies above 640 Hz. For standard ESP operating speeds of 3000 to 4500 RPM the blade passing frequency ranges from 600 to 900 Hz. The aluminum impeller's first mode at 643 Hz sits close to this range and warrants monitoring. Steel and titanium have more margin, with first modes above 699 Hz.

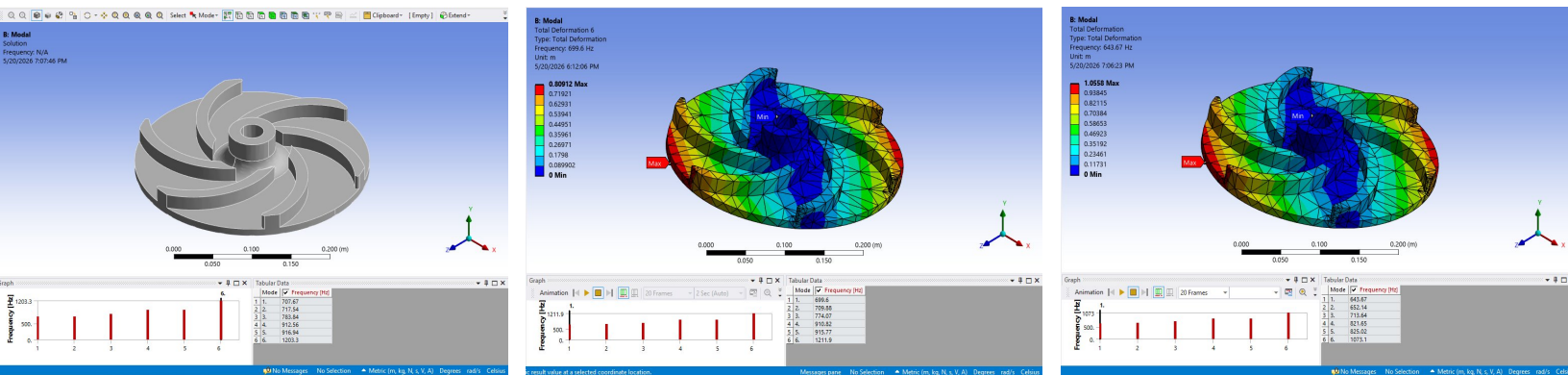


Figure 9. Natural frequency plots for all three materials. Red bars represent individual mode frequencies.

8. Material Comparison and Discussion

Looking across the results, a few clear trends emerge. Stress generally follows centrifugal loading trends and increases rapidly with rotational speed. Steel at 6000 RPM shows roughly 3.5 times the stress of steel at 3000 RPM, close to but not exactly four times, which is expected given that geometry and mesh effects introduce local variations beyond simple centrifugal scaling. Total deformation follows the same trend and increases significantly with speed.

Comparing materials at 9000 RPM, aluminum actually produces lower peak stress than steel because its lower density means less centrifugal force per unit volume even though its stiffness is lower. However, its safety factor of 0.85 is still below 1.0, meaning fatigue failure would occur over time. Titanium at 162.6 MPa sits between the two but has a yield strength of 930 MPa, giving it by far the most structural margin at high speed. For an application where weight matters, like an ESP pump suspended on a cable in a deep well, titanium's strength-to-weight ratio makes it the most attractive choice at high operating speeds.

At 3000 RPM, structural steel shows a safety factor of 2.80 and is entirely adequate for continuous operation. The safety factor drops below 1.0 by 6000 RPM, so the practical operating limit for this steel geometry lies somewhere between those two speeds. Steel is also significantly cheaper than titanium and easier to machine, which is why it dominates real-world pump manufacturing for standard operating conditions.

Material	Density (kg/m3)	Max Stress at 9000 RPM	Safety Factor	Fatigue Life at 9000 RPM	Best Use Case
Steel	7850	277.3 MPa	0.31	8,396 cycles	Safe at 3000 RPM
Aluminum	2770	97.7 MPa	0.85	22M+ cycles	Weight critical, low speed
Titanium	4620	162.6 MPa	High*	N/A	High speed, demanding environment

* Titanium yield strength of 930 MPa provides substantial margin over the 162.6 MPa maximum stress observed at 9000 RPM.

9. Conclusion

This study evaluated a centrifugal pump impeller under realistic centrifugal loading conditions using ANSYS Mechanical. The main findings are summarized below.

1. Structural steel remained safe at 3000 RPM with a safety factor of 2.80. At 6000 RPM and above, the safety factor dropped below 1.0, indicating failure risk. The critical failure threshold lies between 3000 and 6000 RPM and would require additional simulations at intermediate speeds to pinpoint exactly.
2. Aluminum alloy produces lower absolute stress than steel at the same speed due to its lower density, but its lower yield strength means the safety factor at 9000 RPM (0.85) is still below acceptable limits for continuous operation.

3. Titanium alloy shows intermediate stress at 9000 RPM (162.6 MPa) but its high yield strength of 930 MPa gives the best structural margin of the three materials at elevated speeds, making it the preferred choice for high-performance pump applications where weight is also a concern.
4. Modal analysis shows all three materials have first natural frequencies above 640 Hz, which is above the blade passing frequency at 3000 RPM operating speed. No resonance risk was identified within the standard ESP operating envelope.
5. Future work could include a full CFD analysis to apply realistic fluid pressure loads on the blade surfaces, which would provide a more accurate picture of combined hydraulic and centrifugal loading. Mesh independence studies could also be performed to ensure numerical convergence of stress predictions.